Chipsee

Industrial HMI Display

# EPC-ESP32P4-050




PN: CS12720-ESP32P4-050E

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# EPC-ESP32P4-050




Front View



Rear View



Side View 1



Side View 2

## 1 Product Overview

The EPC-ESP32P4-050 Embedded HMI Display (PN: CS12720-ESP32P4-050E) is an embedded programmable capacitive touch HMI display featuring an ESP32 P4 processor. The product features an embedded mounting design, making it suitable for integration into existing equipment.

It features a 5” IPS display with a maximum brightness of 400 cd/m$^2$. The touch screen is a responsive capacitive screen and supports multi-touch.

The product is powered by an ESP32-P4 processor, making it suitable for applications such as:

- Smart Home
- Industrial Automation
- Healthcare
- Consumer Electronics
- Smart Agriculture
- Retail Self-Service Terminals (POS, Vending Machines)
- Service Robots

- Multimedia Player
- High-Speed USB Host and Device
- Smart Voice Interaction Terminal
- Edge Vision AI Processor
- HMI Control Panel

The product can be programmed with ESP-IDF, ESPHome and other ESP32 ecosystem tools and frameworks that support the ESP32-P4.

## 2 Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order EPC-ESP32P4-050 from the official Chipsee Store or from your nearest distributor.

## 3 Hardware Features

The EPC-ESP32P4-050 Embedded HMI Display offers a broad range of performance and connectivity features for flexible integration. Some of the key features are listed below.

**Key Features**

| | |
|---|---|
| **CPU** | Dual-core 32-bit RISC-V, up to 400 MHz; single-core 32-bit LP RISC-V, up to 40 MHz |
| **Storage** | 32MB external SPI Flash |
| **RAM** | 768 KB HP L2MEM + 32 KB LP SRAM + 8 KB SPM; 32MB in-package PSRAM |
| **FRAM** | Optional, not mounted by default |
| **Gyroscope** | Optional, not mounted by default |
| **Display** | 5" IPS LCD, 1280 x 720 px, brightness 400 cd/$m^2$ |
| **Touch** | 5-point capacitive touch with 1.0mm tempered glass |
| **LAN** | 1 x 100Mbps LAN |
| **PoE** | N/A |
| **WiFi/BT** | WiFi 6, Bluetooth LE 5 **(optional, ESP32-C6 is not mounted by default)** |
| **4G/LTE** | Internal Cat 1 4G/LTE module, not mounted by default |
| **Audio** | 3.5mm audio in/out interface; 2W internal speaker |
| **Buzzer** | Onboard buzzer, driven by GPIO |
| **Camera** | 1 x MIPI-CSI (2-lane) interface for connecting a MIPI-CSI (2-lane) 15-pin 1.0mm camera such as a Raspberry Pi (OV5647) Camera V1.3 **(optional, not mounted by default)** |
| **GPIO** | N/A |
| **RS232** | 2 x RS232 **(default)**; up to 4 x RS232 **(optional, by configuring RS485 to RS232)** |

| | |
|---|---|
| **RS485** | 2 x RS485 **(default)**; 2 x RS485 can be configured to 2 x RS232 **(optional)** |
| **CAN** | 1 x CAN **(default)** |
| **USB** | 2 x USB 2.0 Type-A high-speed host ports; 1 x USB 1.1 Type-C port for firmware flashing and debugging; the type-C port supports powering the device while programming |
| **I2C** | 1 x I2C (4 pin header) |
| **Micro SD card socket** | 1 x TF (micro SD) card slot |
| **RTC** | High-accuracy RTC with supercapacitor, can work 1 week after power off **(default)**. High accuracy RTC with lithium coin battery, can work 3 years after power off **(optional, conflict with speaker)**. |
| **Power Input** | 9V to 32V DC |
| **Current** | 200mA max at 15V, 133mA typical at 15V |
| **Power Consumption** | 3W max, 2W typical |
| **Operating Temp.** | From -20°C to +60°C |
| **Dimensions** | 141.2 x 77.6 x 25.8mm |
| **Weight** | 200g |
| **Mounting** | Embedded |
| **Certification** | CE, RoHS |

## 3.1 Optional Features

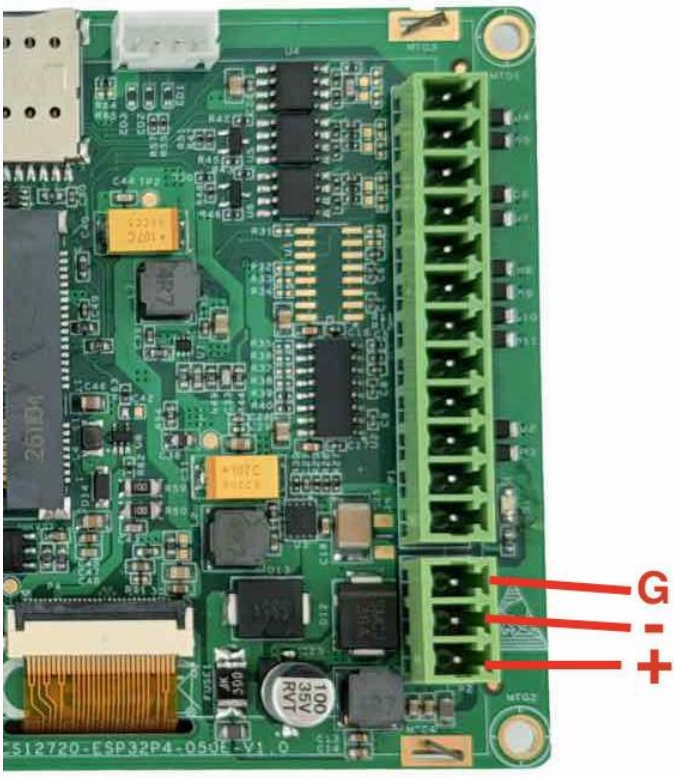
- WiFi/Bluetooth: If Wi-Fi/Bluetooth connectivity is required, an ESP32-C6 module can be installed at the factory. The ESP32-C6 is not populated by default.
- 4G/LTE: 4G/LTE (Cat 1) module is supported on this product, it's not mounted by default, if you need the interface or the 4G/LTE module you can contact us when you place an order.
- Gyroscope is supported but not mounted by default, if you need gyroscope, contact us before placing an order.
- FRAM (Ferroelectric Random Access Memory) is supported but not mounted by default, if you need FRAM, contact us before placing an order.
- 15 pin CSI camera interface is customizable, if you want to use a camera (like Raspberry Pi camera V1.3 OV5647) on the product, contact us.

> ⚠ **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only. Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

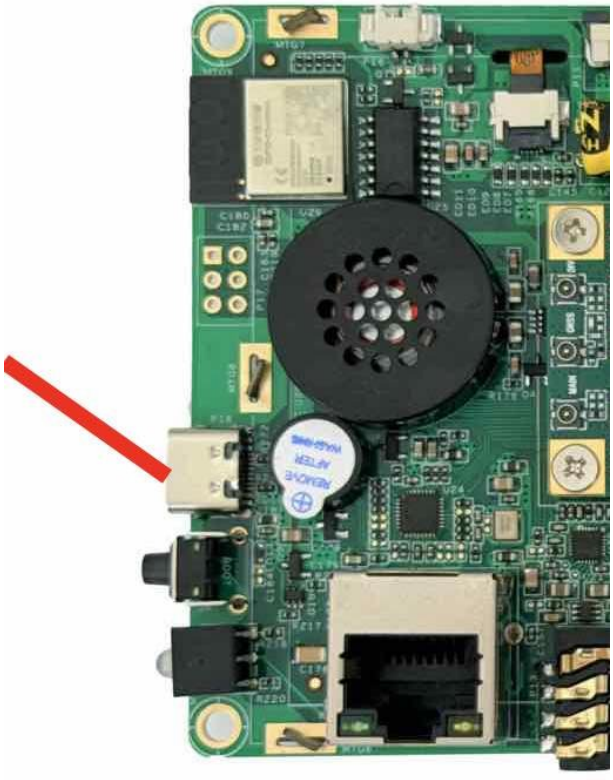
# 4 Power Input

The EPC-ESP32P4-050 Embedded HMI Display can be powered by an input voltage of 9V to 32V DC.

The power input is **3-pin 3.81mm screw terminal**, you can also use the USB-C port to power the product while programming.




Power Connector



USB Power Input

Note that the "**+**" sign represents the positive power input, the "**-**" terminal is shorted to the ground.

**Power Connector**

| Pin Number | Definition | Description |
|---|---|---|
| Pin 1 | Positive Input | DC Power **Positive Terminal** |
| Pin 2 | Negative Input | DC Power **Negative Terminal** |
| Pin 3 | Ground | **Power System Ground** |

> **Note**
>
> The system ground "**G**" is connected to power negative "**-**" on board.
>
> The product doesn't support PoE (Power over Ethernet).

# 5 Connectivity

There are many connectivity options available on the EPC-ESP32P4-050.

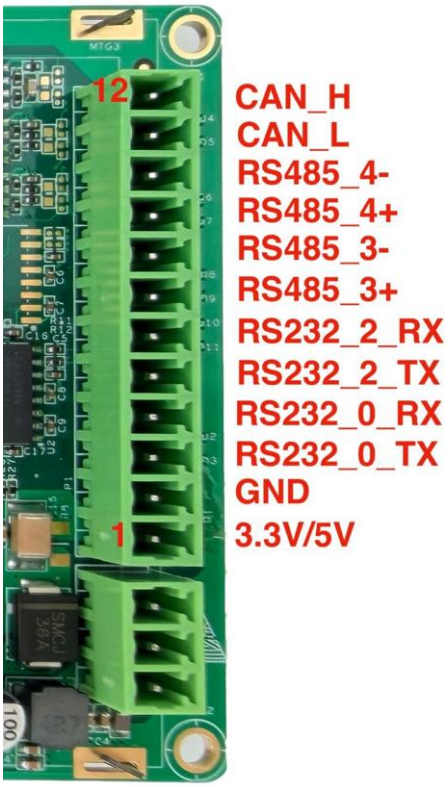
## 5.1 RS232/RS485/CAN

The product provides the following serial and CAN interfaces through a 12-pin connector:

- 2 x RS232 **(default)**; up to 4 x RS232 (**optional**, by configuring RS485 to RS232)
- 2 x RS485 **(default)**; 2 x RS485 can be configured to 2 x RS232 **(optional)**
- 1 x CAN **(default)**

**RS232/RS485/CAN Connector Pin Definition**

| Pin Number | Definition | Description | ESP32 P4 Pin |
|---|---|---|---|
| Pin 12 | CAN_H | CAN high signal | GPIO22 |
| Pin 11 | CAN_L | CAN low signal | GPIO21 |
| Pin 10 | RS485_4- | RS485 –(B) signal | GPIO47 |
| Pin 9 | RS485_4+ | RS485 +(A) signal | GPIO48 |
| Pin 8 | RS485_3- | RS485 –(B) signal | GPIO4 |
| Pin 7 | RS485_3+ | RS485 +(A) signal | GPIO5 |
| Pin 6 | RS232_2_RXD | RS232 RXD signal | GPIO2 |
| Pin 5 | RS232_2_TXD | RS232 TXD signal | GPIO3 |
| Pin 4 | RS232_0_RXD | RS232 RXD signal | GPIO38 |
| Pin 3 | RS232_0_TXD | RS232 TXD signal | GPIO37 |
| Pin 2 | GND | System ground | GND |
| Pin 1 | 3.3V/5V | Power output | 3.3V/5V |




RS232/RS485/CAN Connector

**Attention**

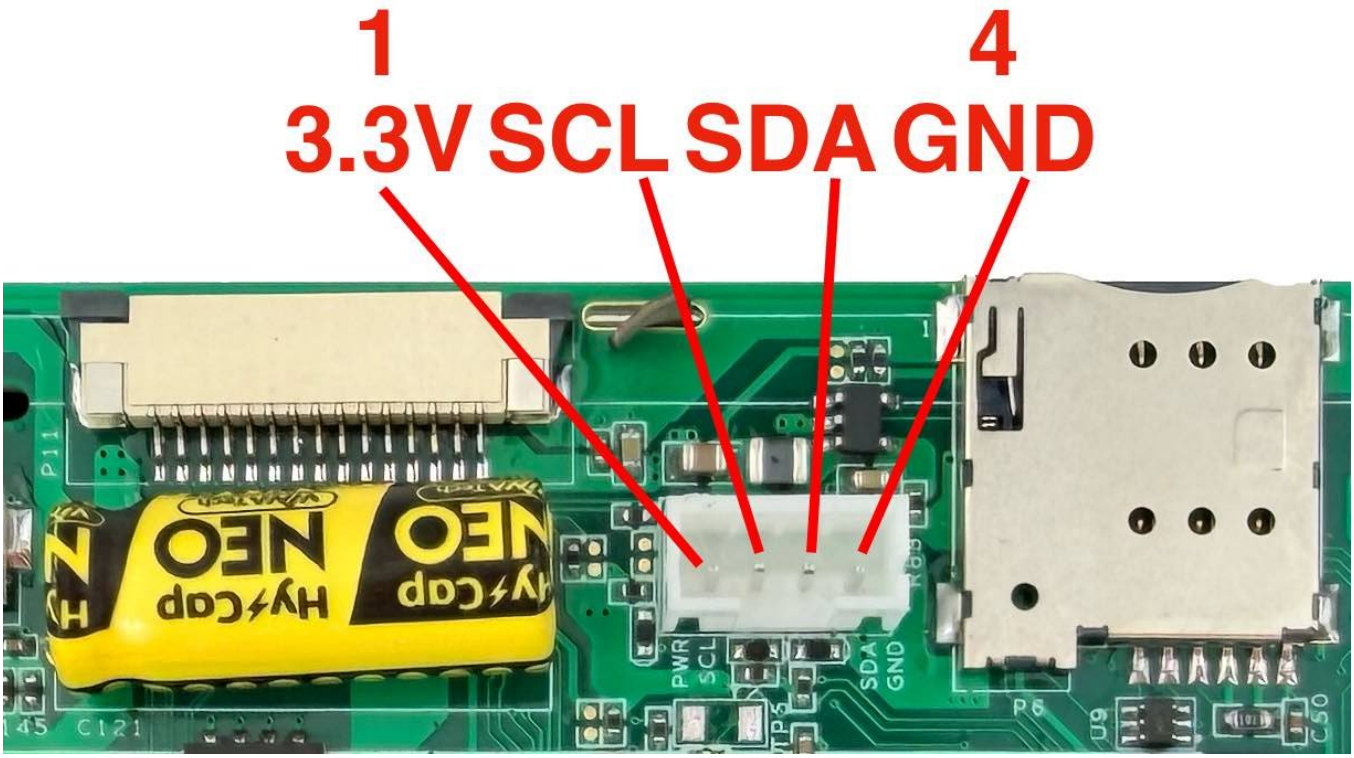
1. Pin 1 is 5V by default, if you need 3.3V, you can **Contact us** before shipping.
2. RS485_3, RS485_4 automatically control input/output direction. They do not require software control.
3. The 120 Ω termination resistor for the RS485 is **not mounted** by default.
4. If you need a different configuration, you can **Contact us** before shipping, we can make a customized version of these pins for you.

## 5.2 I2C

**I2C Connector — 4-pin 2.0mm Pitch Header**

| Pin Number | Definition | Description | P4 |
|---|---|---|---|
| Pin 1 | 3.3V/5V | Power output: 3.3V by default; 5V available as a factory customization | |
| Pin 2 | SCL | I2C clock signal | GPIO8 |
| Pin 3 | SDA | I2C data signal | GPIO7 |
| Pin 4 | GND | System ground | |

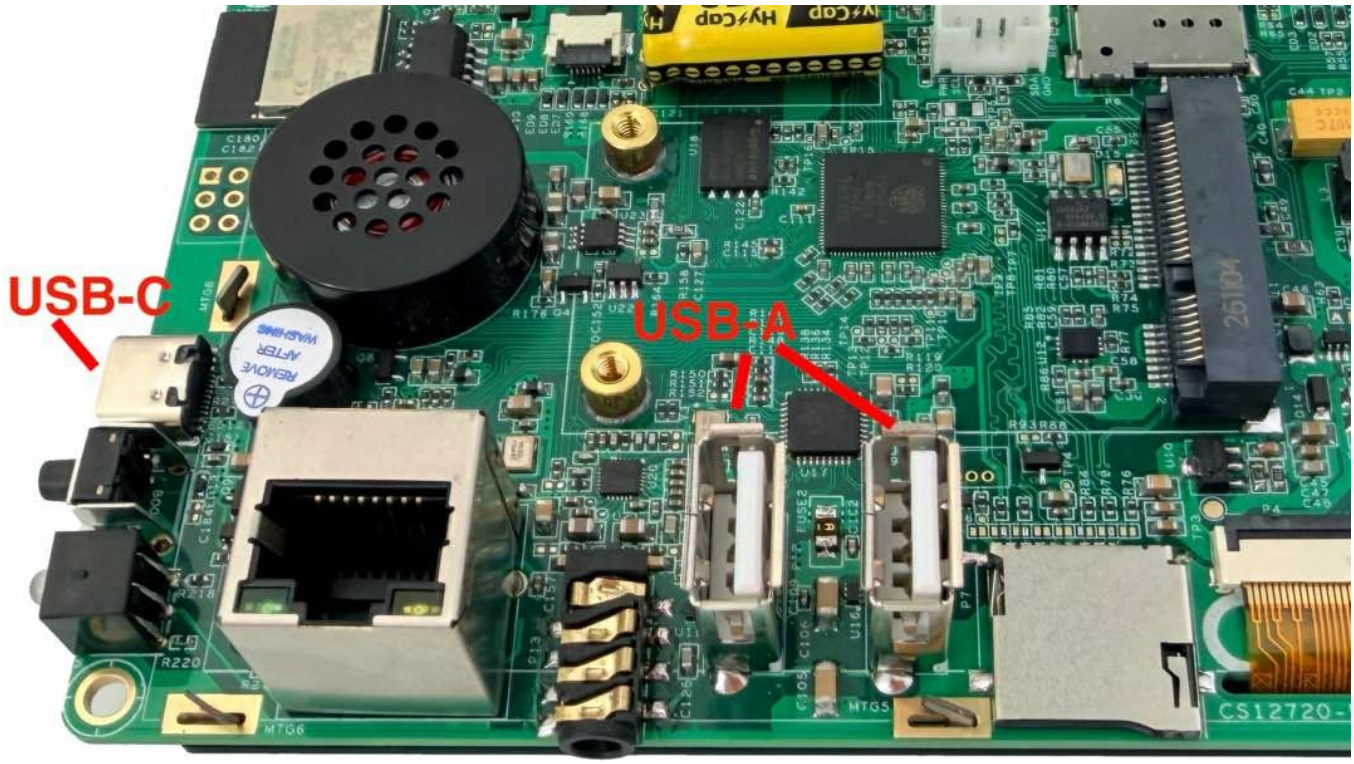
I2C is shared with other I2C devices on board like RTC/touch/audio IC etc. Try to avoid: 0x10 (audio), 0x14/0x5D (touch), 0x32 (RTC), 0x36 (camera, if fitted), 0x50 (FRAM, optional), 0x6A/0x6B (gyroscope, optional)




I2C

## 5.3 USB

There are 2 x **USB2.0 Type-A HOST connectors** and 1 x **Type-C connector** onboard, as shown in the image below.




USB2.0 Host and Type-C Ports

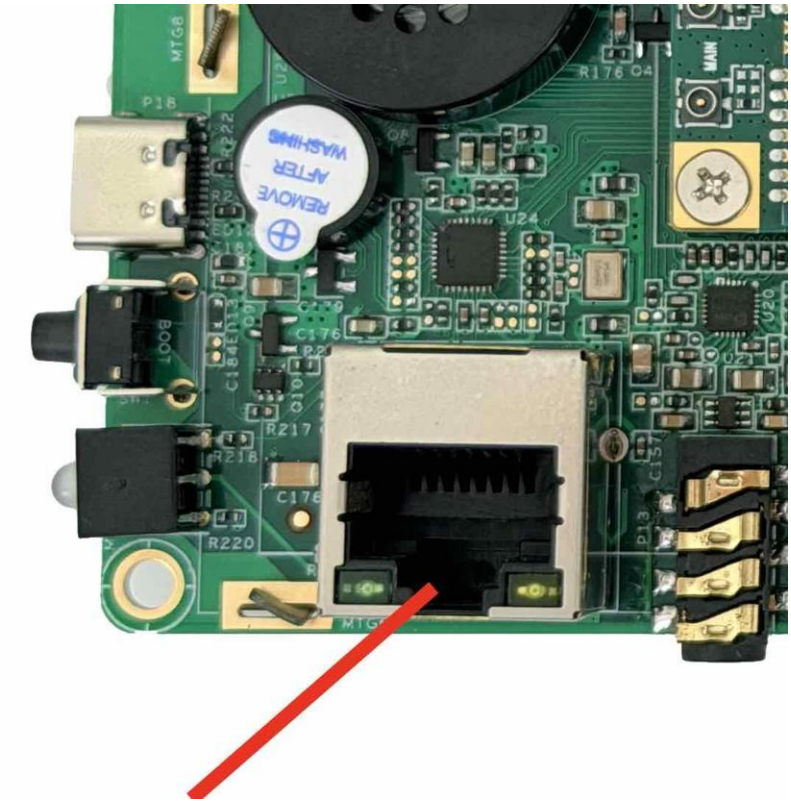
**Note**

- USB peripherals such as high-speed flash drives can be connected to the USB 2.0 host ports.
- Full/Low speed USB devices such as mice and keyboards **cannot** connect due to **the limitation of USB Hub Host software stack**, Espressif managed software stack doesn't support Transaction Translator layer yet.
- The USB Type-C port is used for debugging and firmware flashing and supports powering the product.

## 5.4 LAN

**LAN (RJ45) connector** provides Ethernet connectivity over standardized Ethernet cable. The integrated Ethernet interface supports up to 100 Mbps data throughput.

The product doesn't support PoE (Power over Ethernet).




RJ45 LAN Connector

## 5.5 Wi-Fi and Bluetooth

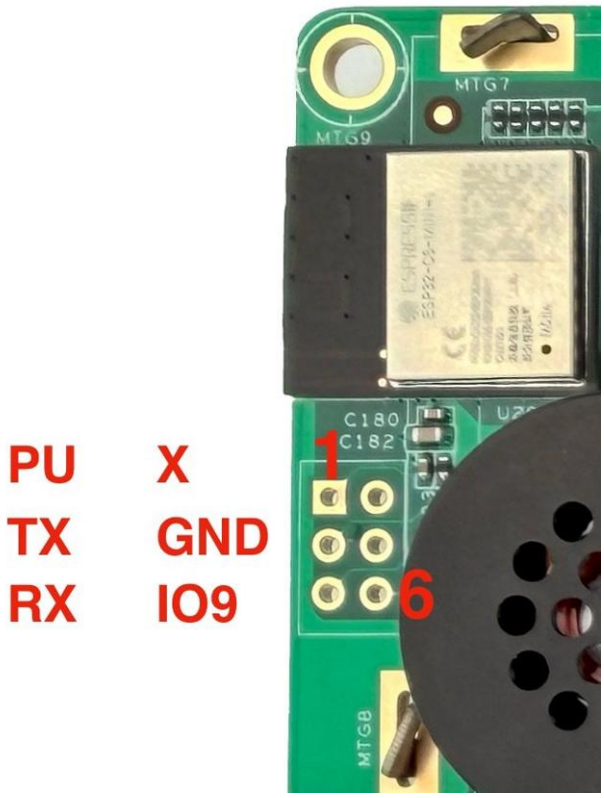
When the optional Wi-Fi/Bluetooth configuration is selected, an ESP32-C6 module provides Wi-Fi and Bluetooth connectivity to the ESP32-P4 through SDIO.

The ESP32-C6 operates as a coprocessor.

Espressif's official ESP-Hosted Wi-Fi and Bluetooth firmware will be flashed at the factory before shipping.

**ESP32-C6 Programming Connector Pin Definition**

| Pin Number | Definition | Description |
|---|---|---|
| Pin 1 | C6_CHIP_PU | ESP32-C6 chip enable/reset signal |
| Pin 2 | X | |
| Pin 3 | C6_U0TXD | ESP32-C6 UART0 transmit signal |
| Pin 4 | GND | System ground |
| Pin 5 | C6_U0RXD | ESP32-C6 UART0 receive signal |
| Pin 6 | C6_IO9 | ESP32-C6 boot mode selection signal |




ESP32-C6 Pin Definition

You can flash the C6 firmware by yourself too. The process is the same as the official Espressif ESP32C6 firmware flashing guide.

The ESP32-C6 firmware can be updated using either of the following methods:

- **UART flashing:**
  1. Prepare a USB-UART (TTL level) cable, to connect the pins and your computer.
  2. Connect GND (Pin 4) to your UART cable's GND, connect GND to C6_IO9 (Pin 6) to let C6 enter flash mode.
  3. Connect C6_U0TXD (Pin 3) to your cable's RX pin.
  4. Connect C6_U0RXD (Pin 5) to your cable's TX pin.
  5. Reset the product/C6 to enter download mode, you can unplug all power sources first and then connect USB-C to your power adapter as a reset.
  6. Remove the C6_IO9 (Pin 6) short to leave flash mode in the next reset.
  7. Reset the product/C6 after flashing to boot normally.

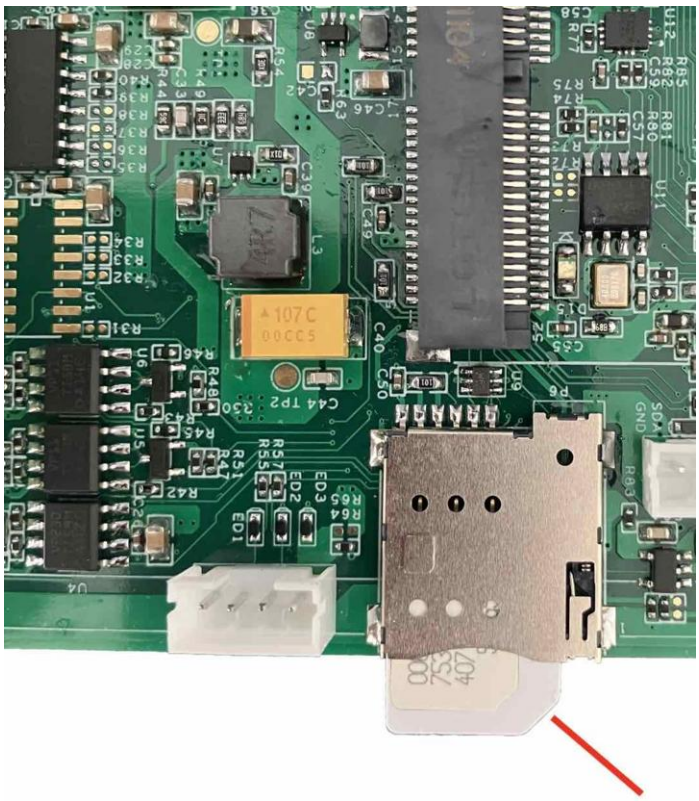
- **Updates through the ESP32-P4:** The ESP32-P4 can flash new firmware to the ESP32-C6 through SDIO using a firmware image if C6 already has a firmware (OTA). This is usually done with a MicroSD card.

## 5.6 4G/LTE

4G/LTE Cat 1 module is **optional**. If you place an order with 4G module included, we will also add a SIM card holder and a 4G/LTE antenna to enable 4G/LTE connectivity.

The SIM card slot does **NOT** support hot-swapping. **Power off** the device before inserting or removing the SIM card.

The Cat 1 module doesn't support GPS.



SIM Card Direction

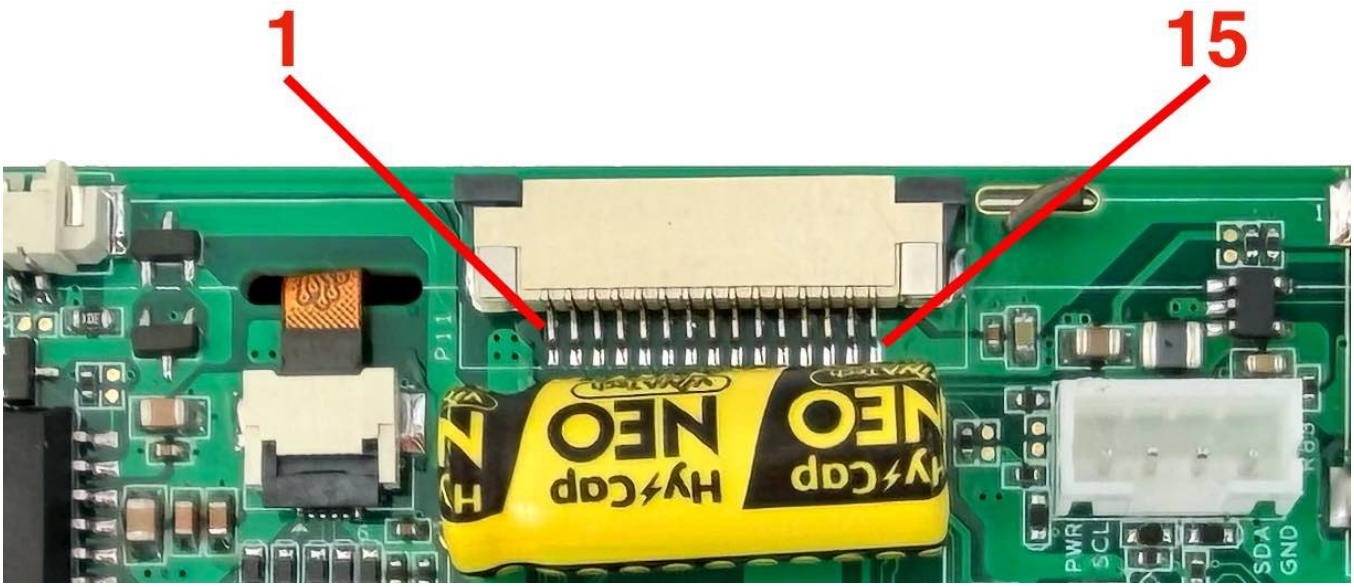
> **Attention**
>
> The product does not come shipped with the 4G/LTE module by default. If you need to use 4G/LTE, you can contact us when placing an order. We can install the necessary hardware for you.

## 5.7 Camera

The product has 1 x MIPI-CSI (2-lane) interface for connecting a MIPI-CSI (2-lane) 15-pin 1.0mm camera such as a Raspberry Pi (OV5647) Camera V1.3 **(optional, not mounted by default)**.

Note: I2C address 0x10 is occupied by audio IC, which conflicts with Raspberry Pi Camera V2/V2.1 (Sony IMX219) module. When you select a camera, please choose carefully because not all electrically compatible camera modules have mature software support for the ESP32-P4.

OV5647 uses I2C address 0x36.




15 pin 1.0mm MIPI-CSI (2-lane) interface

### Camera Connector Pin-out

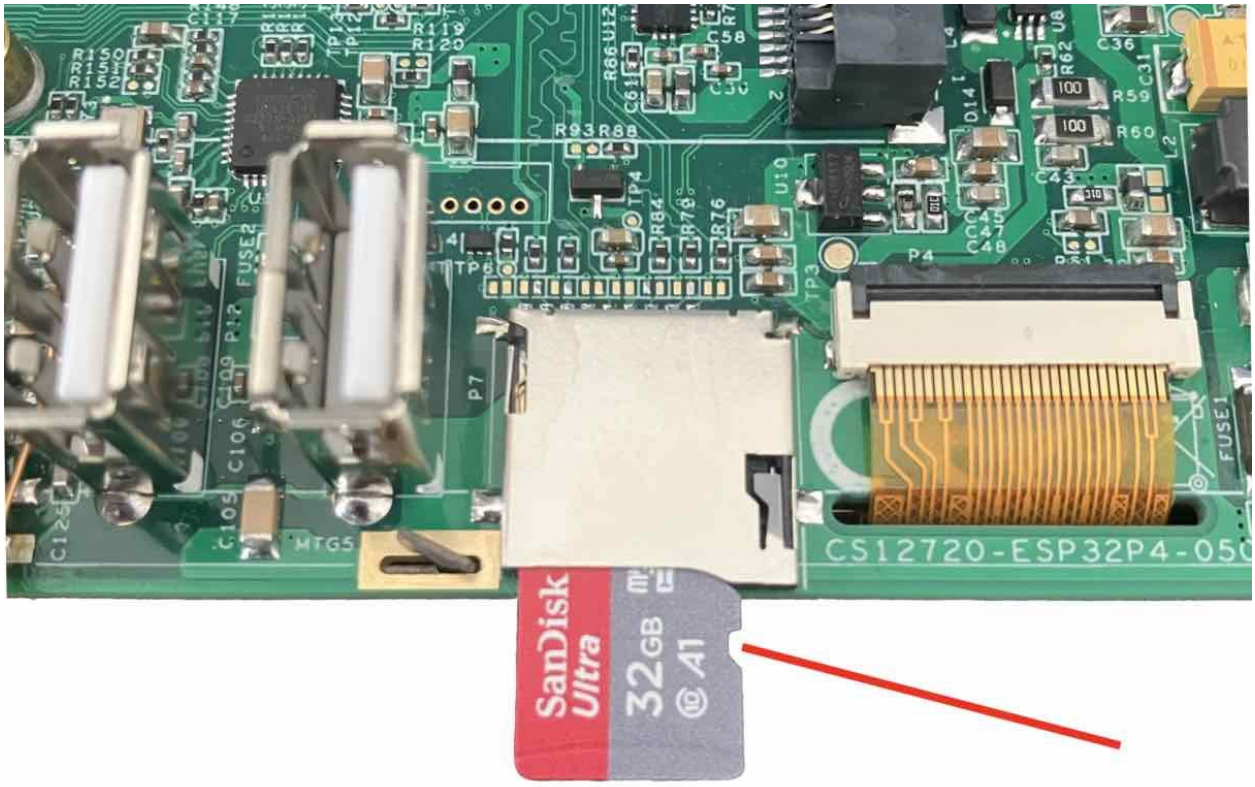
| Camera Connector Pin Definition: | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 1 | GND | Power Ground |
| Pin 2 | CAM1_DN0 | CAM1_DN0 |
| Pin 3 | CAM1_DP0 | CAM1_DP0 |
| Pin 4 | GND | Power Ground |
| Pin 5 | CAM1_DN1 | CAM1_DN1 |
| Pin 6 | CAM1_DP1 | CAM1_DP1 |
| Pin 7 | GND | Power Ground |
| Pin 8 | CAM1_CN | CAM1 Clock signal Negative |
| Pin 9 | CAM1_CP | CAM1 Clock signal Positive |
| Pin 10 | GND | Power Ground |
| Pin 11 | CAM GPIO | Camera enable, pulled up to 3.3V (camera always enabled) |
| Pin 12 | NC | Not connected |
| Pin 13 | SCL0 | I2C SCL (GPIO8), shared with on-board I2C devices |
| Pin 14 | SDA0 | I2C SDA (GPIO7), shared with on-board I2C devices |
| Pin 15 | +3.3V | System +3.3V Power Output, No more than 500mA Current output |

**Attention**

1. The camera interface is not mounted by default.
2. Check the orientation of the ribbon cable (pin 1 / pin 15) carefully, and power off before connecting or removing the camera.

## 5.8 MicroSD Card

The EPC-ESP32P4-050 Embedded HMI Display features 1 x **TF Card (micro SD) slot** (note the direction of the micro SD card). The product does not come shipped with the micro SD card by default.

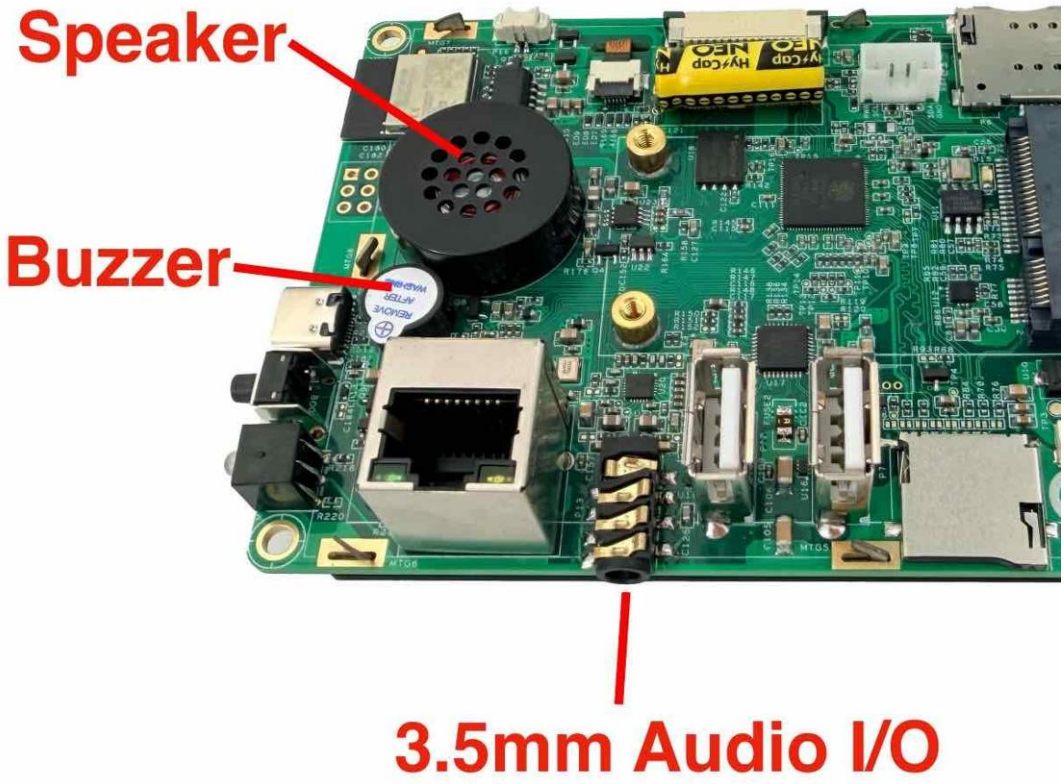



Micro SD Card Slot

## 5.9 Audio

### 5.9.1 3.5 mm Audio In/Out

The product features a 3.5mm audio connector as shown in the image below.




Audio Connector

This connector supports both earphone output and microphone input.

There is no on board microphone for this product.

After inserting a 3.5mm device, the on board speaker will be muted.

### 5.9.2 Internal Speaker

The product also has an internal 2W speaker, meaning you can play music without an earphone, the sound can be played from the product directly.

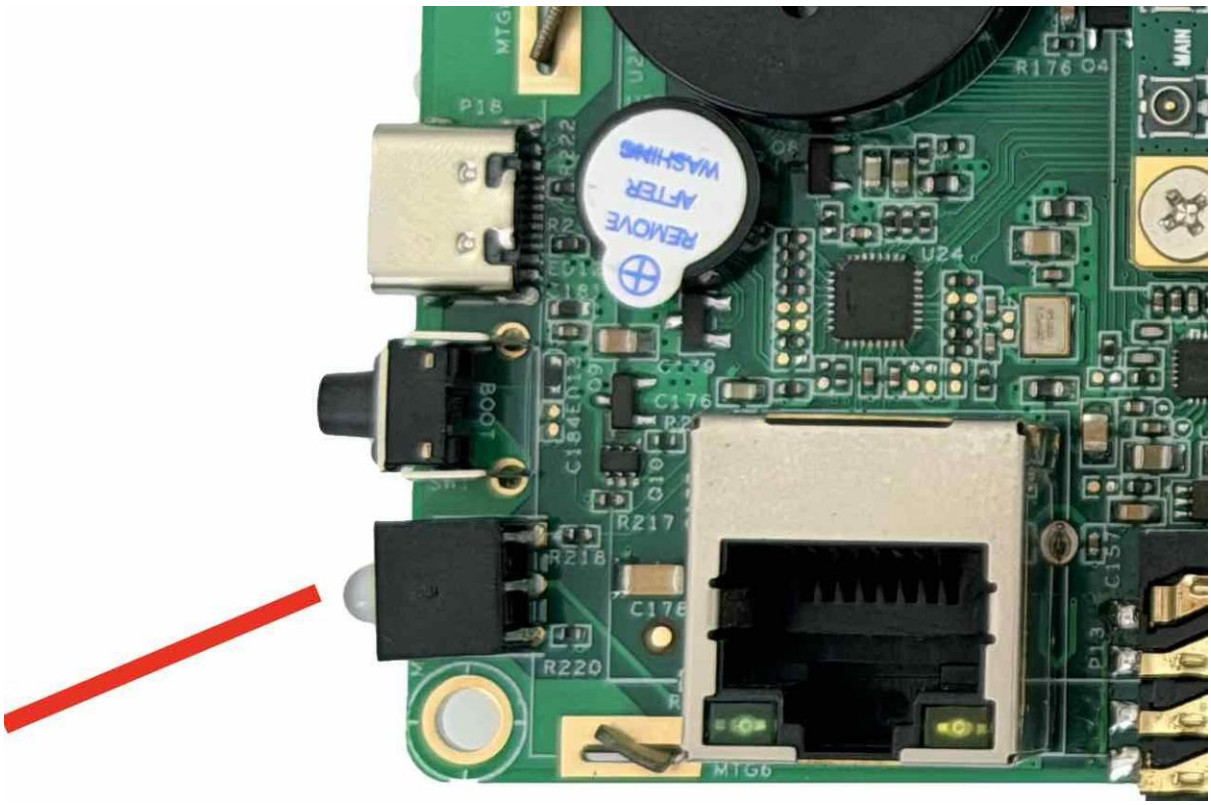
### 5.9.3 Buzzer

The product has an internal buzzer, suitable for playing beep sounds for alarms.

# 6 Button and Status LED

## 6.1 Status LED

This product has a bi-color status LED. It is green whenever the product is powered on. The application can switch it to red through GPIO32 (low = red), for example to signal an alarm or to blink as a heartbeat.

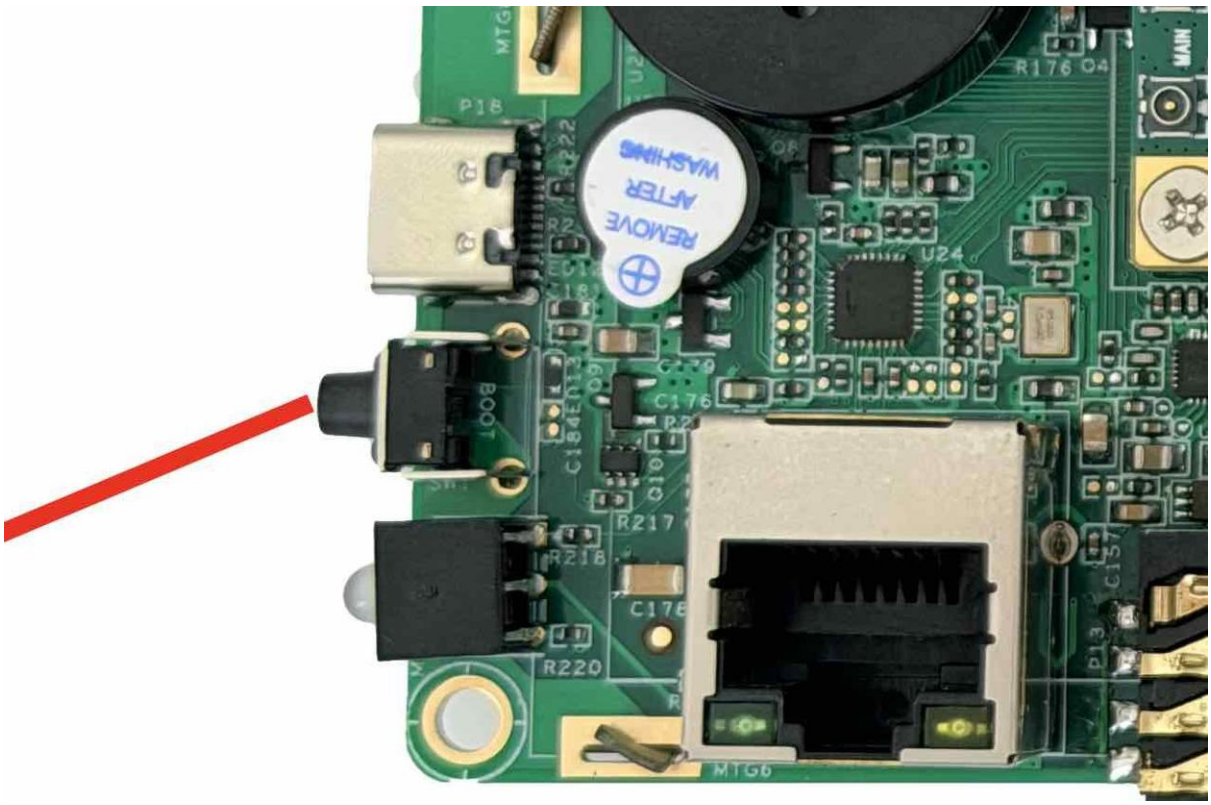



Status LED

## 6.2 Button

There is a PROG button for flashing firmware. Normally when programming with ESP-IDF, the idf software can automatically let the product enter firmware flashing mode.

But if you need to avoid the P4 entering normal mode, you can press and hold the PROG button while powering on or resetting the product. This is helpful when you want to update the C6 firmware and prevent the P4 from communicating with the C6 (WiFi/Bluetooth).




Button

## 7 Mounting Procedure

The EPC-ESP32P4-050 can be installed in an embedded enclosure (guide).

> **Attention**
>
> Do not apply excessive mechanical pressure to the display when mounting it in an enclosure.

## 8 Mechanical Specifications

The outer mechanical dimensions of EPC-ESP32P4-050 are 141.2 x 77.6 x 25.8mm (W x H x D).

## 9 Disclaimer

**This document is provided strictly for informational purposes. Its contents are subject to change without notice. Chipsee assumes no responsibility for any errors that may occur in this document. Furthermore, Chipsee reserves the right to alter the hardware, software, and/or specifications set forth herein at any time without prior notice and undertakes no obligation to update the information contained in this document.**

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# 10 Technical Support

If you encounter any difficulties or have questions related to this document, we encourage you to refer to our other documentation for potential solutions. If you cannot find the solution you're looking for, feel free to contact us. Please email Chipsee Technical Support at **support@chipsee.com**, providing all relevant information. We value your queries and suggestions and are committed to providing you with the assistance you require.